

Joel Hempel

Machine Learning and Fullstack Development

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Summary Statement

Driven software and hardware enthusiast with a strong track record in full-stack development and machine learning. Known for exceeding expectations and swiftly mastering new technologies.

Technologies

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS

Frameworks & Libraries: PyTorch, React, Next.js, FastAPI

Web Development & Databases: Supabase, SQL, Tailwind, PHP

Tools & Environments: Git, Docker, AWS, Google Cloud, JUnit, Google Colab

Machine Learning: Neural Networks, Computer Vision, Language Models

Experience

SWE Intern (ML & Full-Stack Dev) *Metrized Consulting Inc.* **Vancouver, BC** 09/2023 - 08/2024

- Independently lead development of a full-stack application that was deployed within 12-months.
- Held responsibility for improving the object detection (YOLO). Came up with the key feature that unlocked model performance.
- Created the End-to-end/ Integration testing foundation for the application
- Trained, tuned, and optimized neural networks for various tasks.
- Conducted research by reading and implementing findings from academic papers.
- Performed data manipulation, synthetic data generation, and experimental analysis.
- Created multiple presentations for investors and team
- Rigorously documented project of 20 thousand lines

Undergraduate Research Assistant *Systopia Lab* **Vancouver, BC** 05/2023 - 12/2023

- Conducted research using traditional ml-models and deep-learning to locally detect the arrival of spring with satellite and climate data.
- Experimented with data manipulation to create the strongest model within our research.
- Co-authored and virtually presented the conference paper "Flowering Onset Detection: Traditional Learning vs. Deep Learning Performance in a Sparse Label Context" at NeurIPS 2023. [The paper is available here.](#)

IT Intern *Industrial Light & Magic (Lucasfilm)* **Vancouver, BC** 05/2022 - 08/2022

- Built, repaired, and maintained hardware and software for the studio.
- Held the highest ticket completion rate in the team.

Education

BSc Computer Science *University of British Columbia* **Vancouver, Canada** 2022–2025

GPA: 4.00/4.33

Transfer Credits *Langara College* **Vancouver, Canada** 2022–2025

GPA: 4.17/4.33

- Awarded with Ernest E. Livesey Memorial Prize in Mathematics for demonstrating highest achievement in transferable mathematics courses.
- Awarded with LSU Faculty of Science Award for achieving the highest cumulative GPA.